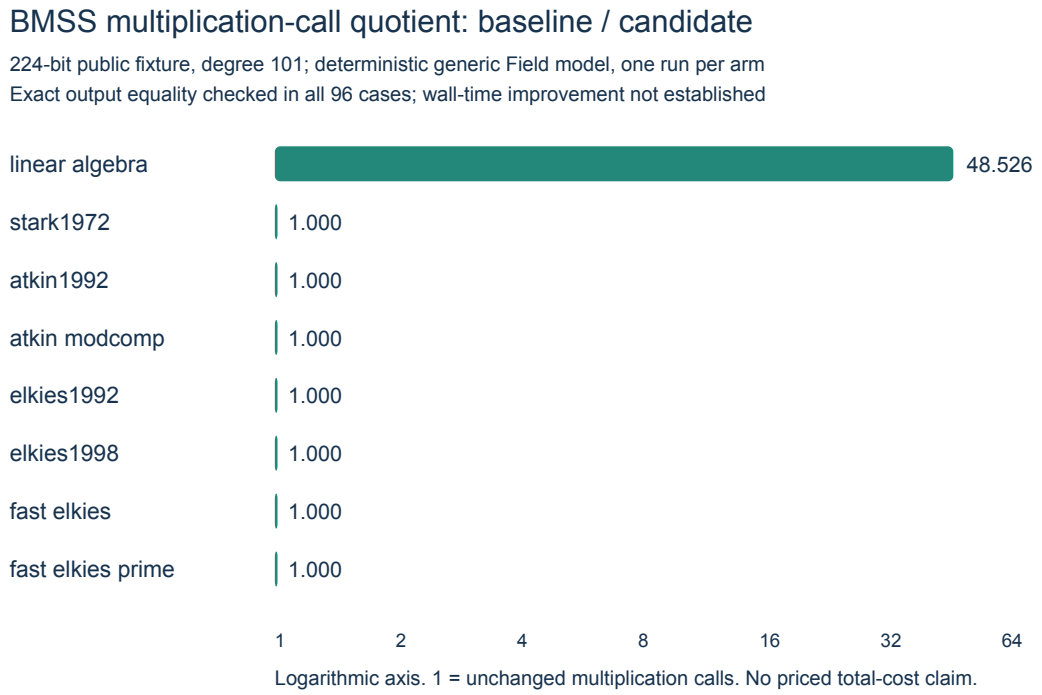
BMSS constructor optimization

2026-10-14 · Native exact-output evidence

All eight BMSS methods at 192, 224, 256 and 511 bits, degrees 7, 31 and 101: 96 exact baseline/candidate output matches. Kernel, codomain and rational map coefficients agree with the Kohel reference. Four public point transports are checked per constructor case.

Arithmetic counts at 224 bits, degree 101

Algorithm	Baseline mul	Candidate mul	Quotient	Baseline inv	Candidate inv
linear algebra	273659435	5639435	48.526	611	611
stark1972	10853630	10853630	1.000	1151	1151
atkin1992	547470	547470	1.000	409	409
atkin modcomp	627414	627414	1.000	417	417
elkies1992	20109	20109	1.000	255	255
elkies1998	11127	11127	1.000	249	249
fast elkies	46958	46958	1.000	108	108
fast elkies prime	1073345	1073345	1.000	110	110



The quotient counts generic Field multiplication calls. Inversions are opaque and counted separately. Setup, preparing sigma, independent reference construction and verification are excluded. Counts have no sampling error for these fixed inputs; their model is not a CPU instruction count or a priced total cost.

Changes and limits

Row-independent series products are computed once per column. The later rung batches fixed series and Newton denominators, with direct division retained at small sizes. Raw receipts distinguish the rungs and retain any additional multiplications incurred by batching.

This shared Apple M4 Pro host has 14 logical cores and 48 GiB RAM. Wall time is unqualified; no wall samples are promoted. Qualified multi-CPU A/A and A/B comparisons, the other inventory families, named NIST whole-search/replay totals, and actual release publication remain open. The 2,000-additional-map search remains separate and unfinished.

Comparison SHA256: 7f5cadad4bc607b456d15fcf60c4363a9243ab28732e44546ad309d883ba42bf Baseline and candidate run digests and complete case inputs are in COMPARISON.json and the source-linked raw JSONL records. Read PROTOCOL.md, INVENTORY.md and ATTEMPTS.md for boundaries and censored attempts.